Chapter-1.

~~There is no difference.~~

1. A host is any device that has an IP address and can send or receive data over a network.

and A end system is a device located at the edge of a network that communicate with other device using the network. In most contexts, hosts and end systems are used interchangeably

~~Example~~;

End system is a broader term - it includes all devices that sit at the network edge (including hosts).

Example of End Systems: Desktops, laptops
Smartphone and Tablets
IoT Sensors
Servers (web server, e-mail server, file server)
Gaming Consoles
Smart TVS

Yes, web server is an end system because it resides at the edge of the network and communicate with other systems (such as client) over the Internet using protocols like HTTP/HTTPS.

2. Standards are very important for network protocol for all devices communicate with each other.

Interoperability:
Standards allow all devices from different companies (like cisco routers, HP servers, Dell PCs) work togather smoothly. Example: A samsung phone can connect to an Apple Wifi router because both follow IEEE Wifi - standards.

Compatibility:
Protocol standards ensure old and new systems can communicate without major change.
Example: Older web browsers can still connect to newer web server using standardized HTTP/HTTPs protocols.

Reliability:
Following a standard ensures that data is transmitted in a predictable and error-free way access networks.

Scalability:
When everyone uses the same protocol standards it is easier to connect new devices to a network...

4. Dial-up modem over telephone line (Home)
DSL over Telephone line (Home)
Cable to HFC : (Home)
4/5G Cellular network: (Wide Area Wireless Access)

100 Mbps switched Ethernet (LAN) ; Enterprise).

5. In hybrid Fiber-Coaxial (HFC) networks:

The down-stream (from ISP to users) transmission rate is shared among users.

The cable modem ~~from~~ termination system (CMTS) at the headend sends data over the shared coaxial cable to many users.

Each user receives all downstream data but only reads the packets addressed to them. The CMTS allocates time slots to avoid collision.

8) Ethernet can run ~~over~~ over ~~set~~ several different types of physical media.

Twisted Pair Copper Cable
CAT-3 (UTP), CAT-5 (UTP), CAT-5e/CAT-6, CAT-6e/CAT 7.

Optical Fible Cable
Multi-mode Fibre
Single Mode Fibre

Coaxial Cable
Thin Coaxial, Thick Coaxial:
Wireless Media
Wireless ethernet (media: Radio waves)

9. HFC : Downstream rate 42.8 Mbps
Upstream rate 30.7 Mbps
bandwidth is shared.

DSL : Downstream rate 24 Mbps
Upstream rate 2.5 Mbps
bandwidth is dedicated.

FTTH: ~~Up~~ Download (10-20) Mbps
Upload (2-10) Mbps
bandwidth is not shared.

| 12. | Circuit-Switched Network | Packet Switched Networks |
|---|---|---|
| Connection Type | Dedicated path is established for the entire communication. | Data is divided into packets, each can take a different route. |
| Delay | Constant, Predictable delay, data rate constant. | Variable delay (depends on congestion and routing) |
| Bandwidth | fixed and guaranteed | shared among ~~use~~ users may fluctuate. |
| Quality | High, suitable for real time communication. | Can experience delay or loss unless managed. |

TDM: Users share the channel in different time slot. ~~Band~~ More bandwidth efficiency. Easier to implement with digital system

FDM: Users share the channel in different frequency bands. Less bandwidth efficiency. Requires analog filters, more complex.

13. (a). Total ~~List~~ link bandwidth = 2 Mbps.

Each users needs 1 Mbps when transmitting

Each user transmit 20% of the time.

Number of users supported $= \frac{2 \text{ Mbps}}{1 \text{ Mbps/user}}$

$= 2$ users.

When circuit switching is used, the link can support 2 users simultaneously.

(b) Only 2 Mbps can be transmitted at a time.

The remaining 1 Mbps of traffic must wait in a queue. Queueing delay occurs.

In packet switching, users share the link ~~St~~ statistically, meaning a link is used only when a user has data to send.

ⓒ Probability that a given user is transmitting is simply:

$$P = 0.2 = 20\%$$

ⓓ Propability that all three users are transmitting simultaneously

$$= \binom{3}{3} p^3 (1-p)^{3-3}$$

$$= (0.2)^3 = 0.008.$$

Query grows is $= 0.008$.

17. Scenario 1: Sender finishes transmitting before the 1st bit reaches the receiver

- Link length: 1000 Km
- Transmission rate: 1 Mbps.
- Packet size = 100 Bytes

Transmission Delay: $\dfrac{\text{Packet size}}{\text{transmission rate}}$

$$= \frac{(100 \times 8)\text{bit}}{1\ \text{Mbps}}$$

$$= 0.0008.$$

Propagation Delay $= \dfrac{\text{Link length}}{\text{propagation speed}}$.

$$= \frac{1000 \text{ km}}{2\times10^{8} \text{ ms}^{-1}}$$

$$= 0.005 \text{ sec}$$

Scenario 2: 1st bit reaches the receiver before the finishes transmitting.

Link length = 100 km.

Transmission rate = 1 Mbps.

packet size = 100 byte.

Transmission delay $= \dfrac{\text{Packet size}}{\text{Transmission Rate}}$

$$= \frac{(100\times8) \text{ bit}}{1 \text{ Mbps}}$$

$$= 0.0008$$

Propagation Delay $= \dfrac{\text{Link length}}{\text{propagation speed}}$.

$$= \frac{100 \text{ km}}{2\times10^{8} \text{ m/s}}.$$

18 packet length = 1000 byte

link size = 2500 km

propagation speed = $2.5 \times 10^8$ m/s.

transmission rate = 2 Mbps.

$$\text{transmission delay} = \frac{\text{packet length}}{\text{transmission rate}}$$

$$= \frac{(1000 \times 8)}{2 \times 10^6}$$

$$= 4 \text{ ms}.$$

$$\text{progation delay} = \frac{\text{link size}}{\text{propagation speed}}$$

$$= \frac{2500 \times 1000}{2.5 \times 10^8}$$

10 ms.

Total time for the last bit to reach the receiver

$$d_{total} = d_{trans} + d_{prop}$$

$$= 4 \text{ ms} + 10 \text{ ms} = 14 \text{ ms}.$$

CamScanner

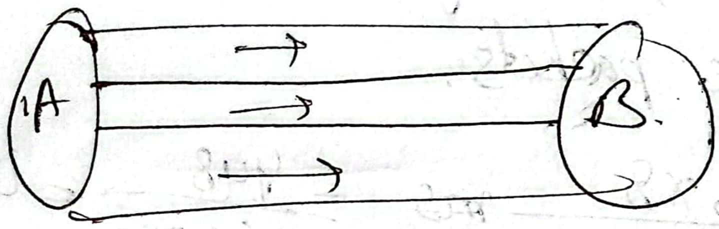
# Throughput : Bottleneck link.

19. a) Throughput = min($R_1$, $R_2$, $R_3$)

= min( 500 kbps, 2 Mbps, 1 Mbps)

= 500 kbps.

(b)




file is 4 million bytes.

file size = ($4 \times 10^6 \times 8$) bit

Through = 500 kbps

= (500×100) bps.

$\frac{L}{R} = \frac{m}{s}$

$\boxed{\frac{L}{R} S = m.}$

Transfer Time = $\frac{\text{File size}}{\text{Throughput}}$

$= \frac{32 \times 10^6}{5 \times 105} = 64$ sec.

(c) Throughput = min($R_1$, $R_2$, $R_3$)

= min( 500 kbps, 100 kbps, 2Mbp

= 100 kbpe.

Transfer Time = $\frac{\text{File size}}{\text{Throughput}} = \frac{4 \times 10^6 \times 8}{100 \times 1000} = 320$ sec.

(P7) Host A converts analog to digital 64 kbps bit stream on the fly.

Host A then groups the bits into 56 byte packets.

Packetization $= \frac{56 \times 8}{64 \times 1000}$ ~~ms~~ $= \frac{448}{64000} = 0.007$ s

$= 7$ ms.

transmission delay $= \frac{56 \times 8}{10 \times 10^6} = \frac{448}{10^7}$

$= 0.045$ ms

~~dt~~ propagation $= 10$ ms.

$(7 + 10 + 0.045)$ ms

$= 17.045$ ms.

$$P = \binom{120}{n} (0.1)^n (0.9)^{120-n}.$$

$$\binom{120}{n} p^n (1-p)^{120-n}.$$

P9. 1 Mbps link.

$$\frac{1 \times 10^{6}\ bps}{\cancel{1000}\ 100 \times 10^{3}\ bps} = \frac{10^{6}}{10^{4}} = 100$$

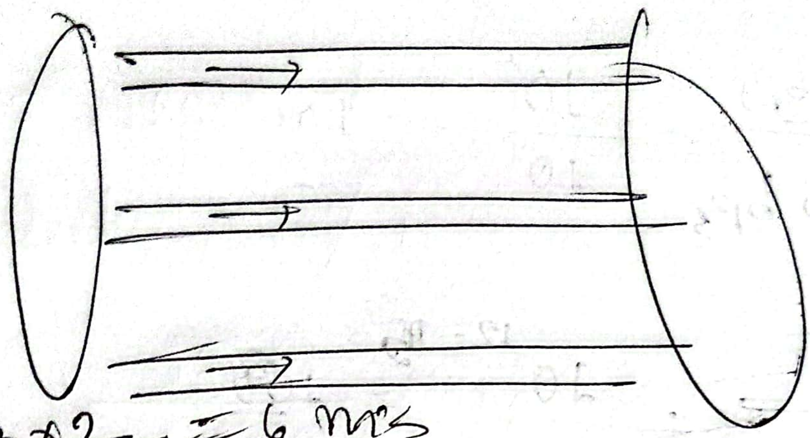
$$\frac{1 \times 10^{12}}{10^{5}} = 10^{12-5} = 10^{7}$$

1 Gbps $= 10^{9}$

$$\frac{10^{9}}{10^{5}} = 10^{4} = 10{,}000\ \text{u sec.}$$

$$\sum_{n=N+1}^{M} \binom{M}{N} p^{n} (1-p)^{M-n}$$

P10. 3 links are



$\cancel{3\times 2} = 6$ ms

$\frac{1500\times 8}{2.5\times 10^6} = 4.8$ ms.

$(4.8\times 3)$ ms $= 14.4$ ms.

$\frac{\cancel{5000}\; 5\times 10^6}{2.5\times 10^8} = 20$ ms.

$\frac{\cancel{4000}\; 4\times 10^6}{2.5\times 10^8} = 16$ ms.

$\frac{\cancel{10\times}\; 10^6}{2.5\times 10^8} = 4$ ms.

End-to-end delay $= (14.4 + 20 + 16 + 4 + 6)$ ms

$= 60.4$ ms.

CamScanner

P.11. Immediately transmit each bit it receives before waiting for the entire packet to arrive. What is the end-to-end delay.

The packet is pipelined across the links. Only one full packet transmission time $L/R$ is required, after that the trailing bit must propagate across each link.

So, end-to-end delay $= \frac{L}{R} + \frac{d_1}{S_1} + \frac{d_2}{S_2} + \frac{d_3}{S_3}$

$= (4.8 + 20 + 16 + 4)$ ms

$= 44.8$ ms.

P.12

$(1500 \times 8)$ bit

$= 12000$ bit

link rate/ transmission rate $= 2.5$ Mbps.

$$\frac{12000}{2.5 \times 10^6} = 4.8 \text{ ms.}$$

Queuing delay $= (4.8 \times 4.5)$ ms $= 21.6$ ms.

13.a)

$$\left(0 + \frac{L}{R} + \frac{2L}{R} + \cdots (n-1)\frac{L}{R}\right) / N.$$

$$= \frac{L}{(R \cdot N)} * (1 + 2 + 3 + \cdots + (n-1)).$$

$$= \frac{L}{RN} * \frac{N(N-1)}{2}$$

$$= \frac{(N-1)L}{2R}$$

Average query delay

L = packet size

R = transmission Rate.

**Problem 16** The total number of packets in the system includes those in the buffer and that is being transmitted, so $N = 10 + 1 = 11$

Because $N = a \cdot d$, so

$$(20+1) = a * (\text{querying delay} + \text{transmission delay}).$$

$$21 = a * \left( 20 \cdot 0.02 + \frac{1}{100} \right)$$

$$N = a \cdot d.$$

$$a = \frac{N}{d} = \frac{100}{0.02 + 0.01} = 3333.33 \text{ packet/sec.}$$

$a$ = average arrival rate.

50 terabyte

transmissim rate = 100 Mbps

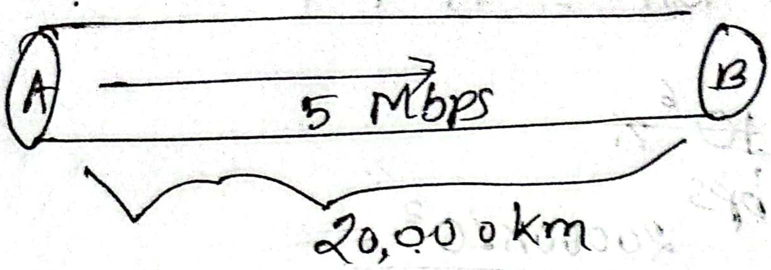
$50 \times 10^{12} \times 8$ bit.

~~46~~

$$\frac{50 \times 10^{12} \times 8}{100 \times 10^{6}} = 4 \times 10^{6} \text{ sec.}$$

$$= 4 \times 10^{6} \text{ Ms.}$$

But with FedEx overnight delivery, you can guarantee the data arrives in one day.




a) Calculate the bandwidth-delay poduct , R.dprop.

$$d_{prop} = \frac{\cancel{20}\ 2\times10^{7}\,m}{2.5\times10^{8}}$$

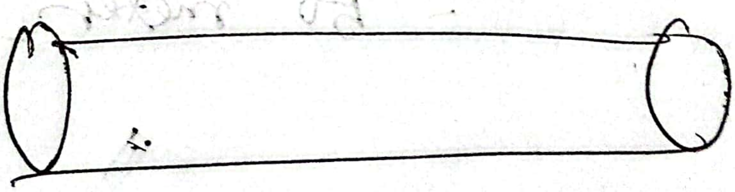
$= 0.08$ sec $= 80$ ms.

R.d = 5 Mbps × 80 ms.

$= (5{,}000000 \times \cancel{80\times1000}\ 0.08)$ bit.

$= 400000$ bits.

(b)



Max

max bit in link = R.dprop.

$= \cancel{5 \times 10^6 \times}$

$R = 5$ Mbps

$$dprop = \frac{20000 \times 10^3}{2.5 \times 10^8}$$

$$= \frac{2 \times 10^7}{2.5 \times 10^8} = 0.08$$

max bit in link = R. dprop.

$= (5 \times 10^6 \times 0.08)$

$= 400000$ bits

$$\text{width of a bit} = \frac{20000 \times 10^3}{400000}$$

$= 50$ meter

~~R~~ P26 : S/R.

end-to-end delay = How long does it take to send the file, assuming it is sent continuously.

36000 km.

Time to send ~~from~~ message from source host to first packet switch = transmission delay.

At what time will the second packet be fully received at the first switch

$= (2 \times t)$

যদি segmenting করে file / message send হয় তবে,

time at with 1st packet is received at the destination host = ~~3~~ $\times$ (transmission delay $\times$ 3) hops

For 800 th packet is received = 15 ms + (800-1) $\times$ 5.